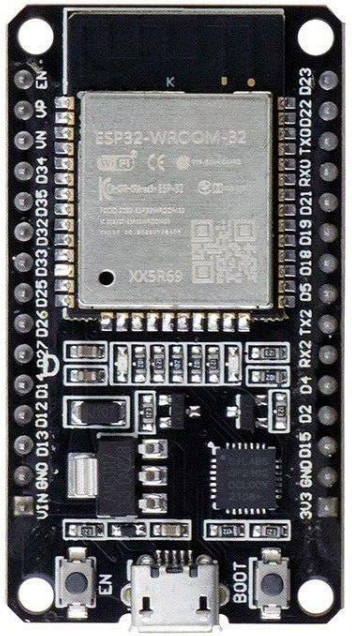
ICT30016 IoT Track – Week 2

ESP32 – The IoT Microcontroller Development Platform

Prepared by Dr. Mark Tee Kit Tsun (SUTS)

Introduction to NodeMCU ESP32



- ESP32 is the name of a microcontroller that has the ability to connect to WIFI (and Bluetooth).

- NodeMCU ESP32 is an electronics development board (like Arduino Uno and M5Stack).

- Important specifications to note about this board:
  - It has WIFI and Bluetooth capability (Important for small IoT applications)
  - Operates on 3.3V (but you can safely power it via USB)
  - Has many pins with various functions (but not all pins can be used)
    - 18 Analog-to-Digital Converter (ADC) channels
    - 3 SPI interfaces
    - 3 UART interfaces
    - 2 I2C interfaces
    - 16 PWM output channels
    - 2 Digital-to-Analog Converters (DAC)
    - 2 I2S interfaces
    - 10 Capacitive sensing GPIOs

Important Task 1:
Study these tutorial pages to learn how to setup with Arduino IDE:
https://randomnerdtutorials.com/getting-started-with-esp32/

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Digital Input & Output using NodeMCU ESP32

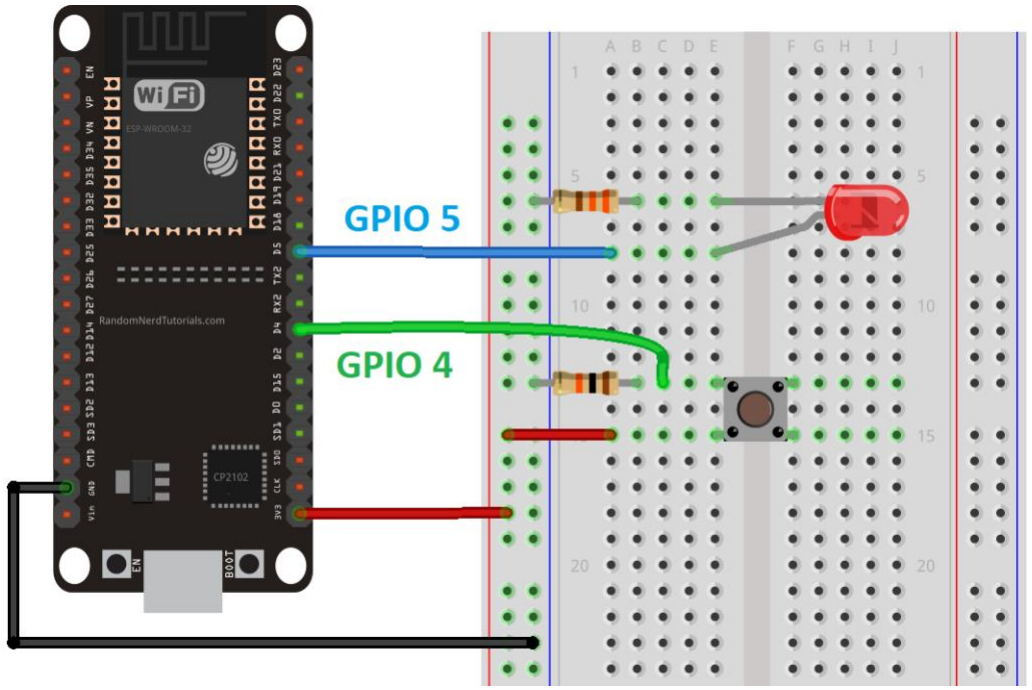
Mastery of the ESP32 begins with knowing how to do:

A. Digital Input (Sensors)

- Reading the pin to see if the connected component is sending a HIGH/LOW signal
- Example applications: Buttons, Switches, IR Sensors, Etc.

B. Digital Output (Actuators)

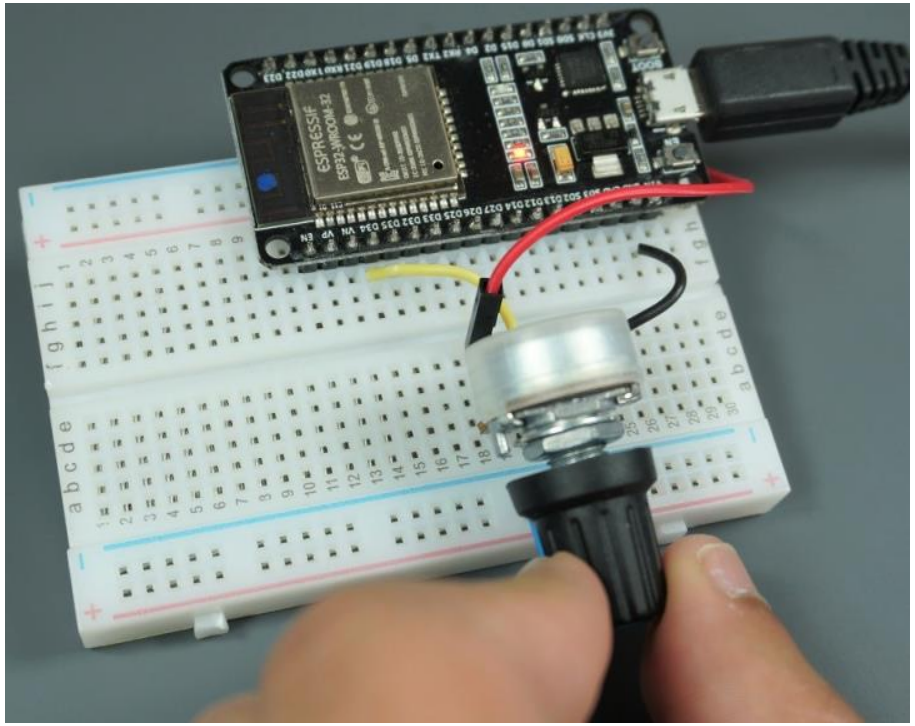
- The ability to output 3.3V (HIGH) or 0V (LOW) voltage from the pin
- Example applications: LED lights, Relay Switches, Etc.



Important Task 2:
Use this Tutorial and the components in your Kit to learn how to utilize LEDs and Buttons with your ESP32:
https://randomnerdtutorials.com/esp32-digital-inputs-outputs-arduino/

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Analog Input using NodeMCU ESP32



- Microcontrollers are essentially digital computers, so they only understand the two binary states: ON or OFF.
  - Equivalent to HIGH/LOW or 3.3V/0V.
  - It does not know how to measure 2.5V or 1.2V.

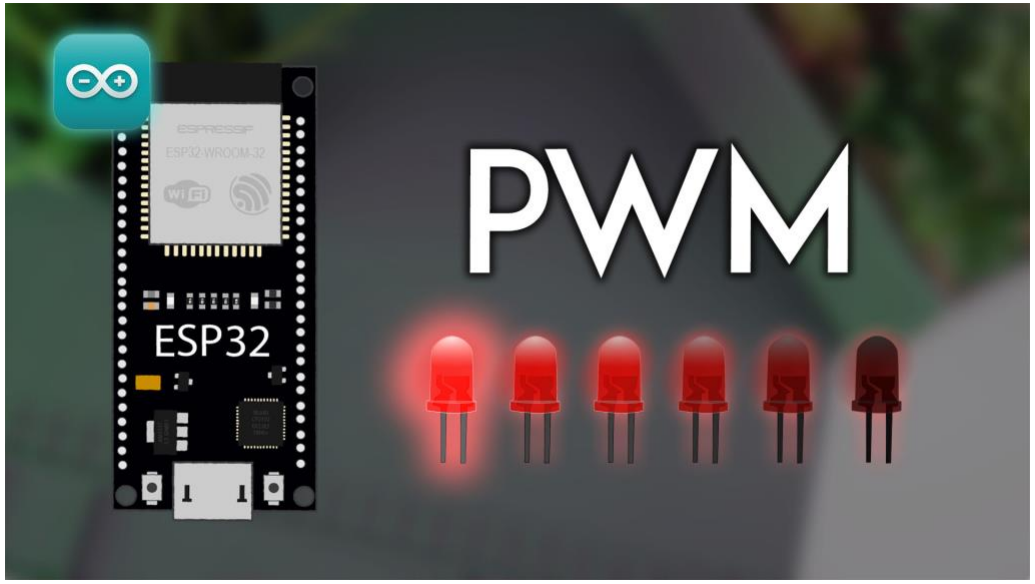
- Then how can ESP32 measure varying voltage levels (AKA Analog Input)?
  - The answer: A sub-system called "ADC"
  - NodeMCU ESP32 has 2 pins capable of ADC.

- Why do you need to master Analog Input?
  - Because a lot of useful sensors send Analog signals.
    - E.g. Temperature, Humidity, Distance, Brightness etc.

Important Task 3:
Use this Tutorial and the components in your Kit to learn how to utilize the Potentiometer with your ESP32:
https://randomnerdtutorials.com/esp32-adc-analog-read-arduino-ide/

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Analog Output using NodeMCU ESP32



- Finally, just like for Analog Input, microcontrollers cannot natively generate signals with varying voltage levels (Analog Output).
- Their pins can either be ON or OFF, but they can use this ability to generate Analog Output.
- This function is called Pulse Width Modulation (PWM).
- PWM is used to control motors, move Servos, dim LEDs and many other functions.

Important Task 4:
Use this Tutorial and the components in your Kit to learn how to utilize PWM for LEDs (as example):
https://randomnerdtutorials.com/esp32-pwm-arduino-ide/

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Takeaway for this Week

1. You must complete Tasks 1-4 in these slides this week to master:
    - Setting up the NodeMCU ESP32 with your Arduino IDE to program it
    - Know how to use LEDs and Buttons as Digital IO
    - Know how to use the Potentiometer as Analog Input
    - Know how to do basic PWM using LED as Analog Output

2. You will notice that the tutorial site (Random Nerd Tutorials) contain a list of 160+ tutorials for projects using the ESP32, including:
    - Guides on how to use basic Digital and Analog Input/Output
    - Advanced functions like Timers, Interrupts, Webservers and More
    - Possible Project Ideas

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